



NCCER Module 22201



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Introduction to Earthmoving





Section Three

Stabilizing Soils

Objective

3. Describe the different methods used in stabilizing soils.
 - a. Identify the different types of binders.
 - b. Explain how to apply binders to soil.
 - c. Explain soil compacting.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

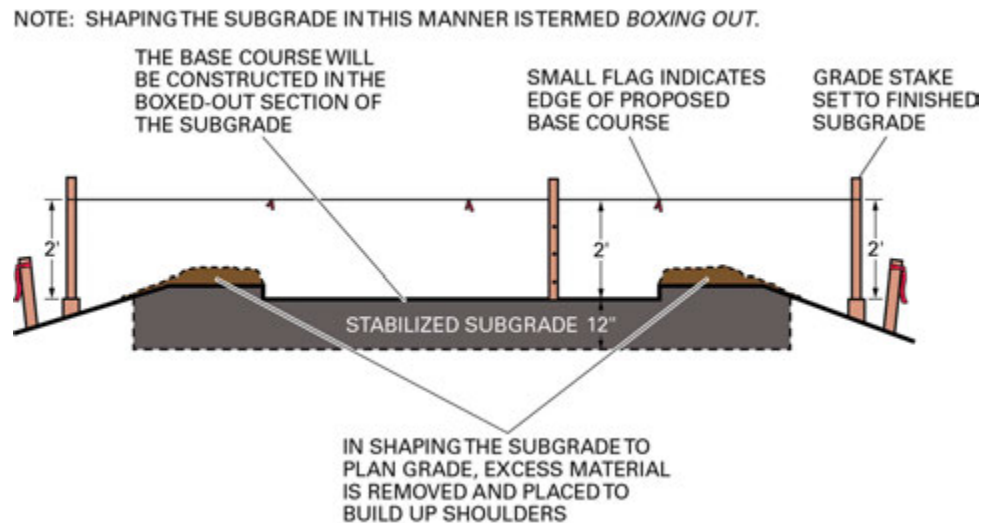
3. Draw a plan for basic earthmoving operations:

- Clearing and grubbing
- Excavating the foundation
- Constructing embankments
- Backfilling
- Compacting



3.0.0 Stabilizing Soils (1 of 3)

- Soil stabilization is the process of preparing sub-base soils to provide a higher loadbearing capacity.
- The process involves pulverizing and mixing the soil thoroughly, often with binders, so that after compaction and curing, the soil is denser, more stable, and stronger.



3.0.0 Stabilizing Soils (2 of 3)

- A soil's stability and loadbearing capacity depend on its cohesion and **shear strength**.
- The higher the soil's shear strength and cohesion, the greater its capacity to bear loads.
- First determine soil properties, the amount and type of binder required, and the degree of compaction needed.



3.0.0 Stabilizing Soils (3 of 3)

- A soil stabilizer machine cuts, mixes, and pulverizes material to modify and stabilize the soil.
- Stabilizing agents may be mixed into the soil by the machine.
- Stabilizers use blades on rotors or teeth on mandrels for cutting. Many have automatic depth control. Some models have a mixing chamber.
- To carry out stabilization by mechanical methods, the entire area must be mixed thoroughly to the designated limits.
- The ability of the machine to accomplish the specified mixing depth is critical to permit one-pass operation.



3.1.0 Choosing a Binder

Binders are selected and added according to:

- The soil type
- The size or gradation of soil particles
- Soil moisture content
- Loadbearing specifications of the project



3.1.1 Cement

- Fine, powdered hydraulic binder
- Calcified lime compounds mixed with silica, alumina, and iron oxide
- Sets and hardens in air or under water
- Provides the highest loadbearing capacity of all binders
- Used in gravel, sand, or reclaimed asphalt
- Can stabilize silt and clay



3.1.2 Lime

- Lime is effective as a binder in clay-bearing soils and aggregates because it reacts both chemically and physically with the soil.
- Quicklime (calcium oxide) and hydrated lime (calcium hydroxide) are used for stabilization.
- Lime reacts chemically with clay soils to produce a cementing action.
- Lime treatment is used for sub-base stabilization, base stabilization, and lime modification.

3.1.3 Fly Ash

- If lime is used on low **plasticity index** soil, a second additive may be used to produce the required reaction to the lime.
- Fly ash is a residue recovered from smoke produced by coal-fired burners that consists of glass, crystalline matter, and carbon.
- It is similar to Portland cement without lime.
- It reacts chemically with lime to enhance the strength of the soil, giving it cement-like properties.
- It can be combined with lime to produce an excellent road base when mixed with aggregate.



3.1.4 Calcium Chloride

- Calcium chloride increases soil density and lowers the freezing point of water.
- The roads that use it for stabilization of the base material are better protected against frost damage.
- It may require less compaction.



3.1.5 Polymers

- In soil stabilization, **polymers** bind with soil particles to create a durable surface that permits motorized traffic.
- Most products are applied as a viscous liquid that dries quickly (in a matter of minutes to hours, depending on the weather).
- Some polymers can be used in a weaker solution to suppress dust for erosion control.



3.2.0 Applying Binders to Soil

- To prepare the subgrade, topsoil and large stones are removed from the proposed site and the subgrade is carefully shaped to grade.
- Grading is done by a motor grader or by a fine grade trimmer. A trimmer creates an accurate profile and can accomplish the job in one pass.
- When grading is complete, scarifying, or disc harrowing to the depth of stabilization is sometimes used.
- Dry binders must be evenly distributed over the area to be stabilized to assure that the specified soil-to-binder ratios are met.
- Lime slurry or another liquid binder is best distributed by tanker truck, which spreads the product to pre-calculated depths as called for in the specifications.



3.2.1 Logistics

Planning considerations must include:

- Atmospheric conditions
- Water supply
- Compaction equipment
- Trimmer equipment



3.3.0 Compacting the Soil (1 of 3)

- The stabilized area must be compacted after the stabilizer has finished.
- If cement is used, the compaction must be done immediately after mixing with vibratory roller.



3.3.0 Compacting the Soil (2 of 3)

- When compacting a lime-stabilized area, the compaction must be done when the moisture content has dropped to a point within the specified range.
- The number of passes made by the compactor depends on the material moisture content, the layer thickness, the compactor type, and the degree of compaction called for in the specifications.
- Various types of equipment use pressure, kneading, vibration, and impaction, or a combination of these methods to compact the soil.



3.3.0 Compacting the Soil (3 of 3)

Each stabilized layer of the roadbed must be checked for:

- Compaction or density
- Thickness
- Mix uniformity
- Gradation
- Loadbearing capacity
- Moisture content
- Binder content



Wrap Up – Trade Terms

Plasticity index: The range of percent water content in which a soil displays plastic or pliable properties. Soils with high plasticity indexes are usually clays; soils with lower indexes are usually silt, sand, or gravel.

Polymers: Natural and synthetic materials made of long molecules containing many repeating chemical subunits. Familiar polymers include plastics and natural rubber. Soil stabilization polymers include acrylic materials.

Shear strength: A property of solid materials that measures its resistance to deforming along planes, called shearing, like spreading a deck of cards. In soil mechanics, it is measured by the angle of internal friction, which relates to the maximum steepness of a soil embankment before it slides due to gravity.



3.0.0 Section Review Question 1

1. According to the definition of soil shear strength, soils with higher shear strengths _____.
 - a. compact easier
 - b. are less stable
 - c. cannot form steep-sided slopes
 - d. do not slide easily down an embankment



3.0.0 Section Review Question 1 Answer

1. According to the definition of soil shear strength, soils with higher shear strengths _____.
- a. compact easier
 - b. are less stable
 - c. cannot form steep-sided slopes
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3.0.0 Section Review Question 2

2. Which of the following statements about the logistics of soil compaction is true?
- a. Soil compaction works best with long, gentle precipitation.
 - b. Soil compaction also shapes the grade to its specified profile and crown.
 - c. Compaction occurs before soil stabilization to provide a firm base.
 - d. Surface trimming shapes the compacted surface to accurate grade and crown specifications.



3.0.0 Section Review Question 2 Answer

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 - d. Surface trimming shapes the compacted surface to accurate grade and crown specifications.**



3.0.0 Section Review Question 3

3. Which of the following is NOT a factor in determining the number of soil compaction passes required?
- a. Amount of compaction required
 - b. Final layer thickness
 - c. Initial soil density
 - d. Moisture content



3.0.0 Section Review Question 3 Answer

3. Which of the following is NOT a factor in determining the number of soil compaction passes required?
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 - b. Final layer thickness
 - c. Initial soil density**
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Next...



Module Review and Examination

Complete the Review Questions at the end of the module and prepare for the Module Exam.